We can use this characterization of arbitrage-free prices and values to solve several interesting problems.

Suppose, for example, that the values V are known, and all prices except the last one, p_n, are known. The set of prices p_n that are consistent with the no-arbitrage assumption is an interval, which can be found by solving a pair of LPs. The optimal value of the LP

$$\begin{array}{ll} \text{minimize} & p_n \\ \text{subject to} & V^T y = p, \quad y \succeq 0, \end{array}$$

with variables p_n and y, gives the smallest possible arbitrage-free price for asset n. Solving the same LP with maximization instead of minimization yields the largest possible price for asset n. If the two values are equal, *i.e.*, the no-arbitrage assumption leads us to a unique price for asset n, we say the market is *complete*. For an example, see exercise 5.38.

This method can be used to find bounds on the price of a derivative or option that is based on the final value of other underlying assets, *i.e.*, when the value or payoff of asset n is a function of the values of the other assets.

5.9 Generalized inequalities

In this section we examine how Lagrange duality extends to a problem with generalized inequality constraints

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \preceq_{K_i} 0, \quad i = 1, \ldots, m \\ & h_i(x) = 0, \quad i = 1, \ldots, p, \end{array} \tag{5.91}$$

where $K_i \subseteq \mathbf{R}^{k_i}$ are proper cones. For now, we do not assume convexity of the problem (5.91). We assume the domain of (5.91), $\mathcal{D} = \bigcap_{i=0}^m \mathbf{dom}\, f_i \cap \bigcap_{i=1}^p \mathbf{dom}\, h_i$, is nonempty.

5.9.1 The Lagrange dual

With each generalized inequality $f_i(x) \preceq_{K_i} 0$ in (5.91) we associate a Lagrange multiplier *vector* $\lambda_i \in \mathbf{R}^{k_i}$ and define the associated Lagrangian as

$$L(x, \lambda, \nu) = f_0(x) + \lambda_1^T f_1(x) + \cdots + \lambda_m^T f_m(x) + \nu_1 h_1(x) + \cdots + \nu_p h_p(x),$$

where $\lambda = (\lambda_1, \ldots, \lambda_m)$ and $\nu = (\nu_1, \ldots, \nu_p)$. The dual function is defined exactly as in a problem with scalar inequalities:

$$g(\lambda, \nu) = \inf_{x \in \mathcal{D}} L(x, \lambda, \nu) = \inf_{x \in \mathcal{D}} \left(f_0(x) + \sum_{i=1}^m \lambda_i^T f_i(x) + \sum_{i=1}^p \nu_i h_i(x) \right).$$

Since the Lagrangian is affine in the dual variables (λ, ν), and the dual function is a pointwise infimum of the Lagrangian, the dual function is concave.

As in a problem with scalar inequalities, the dual function gives lower bounds on $p^\star$, the optimal value of the primal problem (5.91). For a problem with scalar inequalities, we require $\lambda_i \geq 0$. Here the nonnegativity requirement on the dual variables is replaced by the condition

$$\lambda_i \succeq_{K_i^*} 0, \quad i = 1, \ldots, m,$$

where K_i^* denotes the dual cone of K_i. In other words, the Lagrange multipliers associated with inequalities must be *dual* nonnegative.

Weak duality follows immediately from the definition of dual cone. If $\lambda_i \succeq_{K_i^*} 0$ and $f_i(\tilde{x}) \preceq_{K_i} 0$, then $\lambda_i^T f_i(\tilde{x}) \leq 0$. Therefore for any primal feasible point $\tilde{x}$ and any $\lambda_i \succeq_{K_i^*} 0$, we have

$$f_0(\tilde{x}) + \sum_{i=1}^m \lambda_i^T f_i(\tilde{x}) + \sum_{i=1}^p \nu_i h_i(\tilde{x}) \leq f_0(\tilde{x}).$$

Taking the infimum over $\tilde{x}$ yields $g(\lambda, \nu) \leq p^\star$.

The Lagrange dual optimization problem is

$$\begin{array}{ll} \text{maximize} & g(\lambda, \nu) \\ \text{subject to} & \lambda_i \succeq_{K_i^*} 0, \quad i = 1, \ldots, m. \end{array} \tag{5.92}$$

We always have *weak duality*, *i.e.*, $d^\star \leq p^\star$, where $d^\star$ denotes the optimal value of the dual problem (5.92), whether or not the primal problem (5.91) is convex.

Slater's condition and strong duality

As might be expected, *strong* duality ($d^\star = p^\star$) holds when the primal problem is convex and satisfies an appropriate constraint qualification. For example, a generalized version of Slater's condition for the problem

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \preceq_{K_i} 0, \quad i = 1, \ldots, m \\ & Ax = b, \end{array}$$

where f_0 is convex and f_i is K_i-convex, is that there exists an $x \in \mathbf{relint}\, \mathcal{D}$ with $Ax = b$ and $f_i(x) \prec_{K_i} 0$, $i = 1, \ldots, m$. This condition implies strong duality (and also, that the dual optimum is attained).

Example 5.11 *Lagrange dual of semidefinite program.* We consider a semidefinite program in inequality form,

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & x_1 F_1 + \cdots + x_n F_n + G \preceq 0 \end{array} \tag{5.93}$$

where $F_1, \ldots, F_n, G \in \mathbf{S}^k$. (Here f_1 is affine, and K_1 is $\mathbf{S}_+^k$, the positive semidefinite cone.)

We associate with the constraint a dual variable or multiplier $Z \in \mathbf{S}^k$, so the Lagrangian is

$$\begin{array}{rcl} L(x, Z) & = & c^T x + \mathbf{tr}\left((x_1 F_1 + \cdots + x_n F_n + G)\, Z\right) \\ & = & x_1(c_1 + \mathbf{tr}(F_1 Z)) + \cdots + x_n(c_n + \mathbf{tr}(F_n Z)) + \mathbf{tr}(GZ), \end{array}$$

which is affine in x. The dual function is given by

$$g(Z) = \inf_x L(x, Z) = \begin{cases} \mathbf{tr}(GZ) & \mathbf{tr}(F_i Z) + c_i = 0, \quad i = 1, \ldots, n \\ -\infty & \text{otherwise.} \end{cases}$$

The dual problem can therefore be expressed as

$$\begin{array}{ll} \text{maximize} & \mathbf{tr}(GZ) \\ \text{subject to} & \mathbf{tr}(F_i Z) + c_i = 0, \quad i = 1, \ldots, n \\ & Z \succeq 0. \end{array}$$

(We use the fact that $\mathbf{S}^k_+$ is self-dual, *i.e.*, $(\mathbf{S}^k_+)^* = \mathbf{S}^k_+$; see §2.6.)

Strong duality obtains if the semidefinite program (5.93) is strictly feasible, *i.e.*, there exists an x with

$$x_1 F_1 + \cdots + x_n F_n + G \prec 0.$$

Example 5.12 *Lagrange dual of cone program in standard form.* We consider the cone program

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Ax = b \\ & x \succeq_K 0, \end{array}$$

where $A \in \mathbf{R}^{m \times n}$, $b \in \mathbf{R}^m$, and $K \subseteq \mathbf{R}^n$ is a proper cone. We associate with the equality constraint a multiplier $\nu \in \mathbf{R}^m$, and with the nonnegativity constraint a multiplier $\lambda \in \mathbf{R}^n$. The Lagrangian is

$$L(x, \lambda, \nu) = c^T x - \lambda^T x + \nu^T (Ax - b),$$

so the dual function is

$$g(\lambda, \nu) = \inf_x L(x, \lambda, \nu) = \begin{cases} -b^T \nu & A^T \nu - \lambda + c = 0 \\ -\infty & \text{otherwise.} \end{cases}$$

The dual problem can be expressed as

$$\begin{array}{ll} \text{maximize} & -b^T \nu \\ \text{subject to} & A^T \nu + c = \lambda \\ & \lambda \succeq_{K^*} 0. \end{array}$$

By eliminating λ and defining $y = -\nu$, this problem can be simplified to

$$\begin{array}{ll} \text{maximize} & b^T y \\ \text{subject to} & A^T y \preceq_{K^*} c, \end{array}$$

which is a cone program in inequality form, involving the dual generalized inequality.

Strong duality obtains if the Slater condition holds, *i.e.*, there is an $x \succ_K 0$ with $Ax = b$.

5.9.2 Optimality conditions

The optimality conditions of §5.5 are readily extended to problems with generalized inequalities. We first derive the complementary slackness conditions.

Complementary slackness

Assume that the primal and dual optimal values are equal, and attained at the optimal points $x^\star$, $\lambda^\star$, $\nu^\star$. As in §5.5.2, the complementary slackness conditions follow directly from the equality $f_0(x^\star) = g(\lambda^\star, \nu^\star)$, along with the definition of g. We have

$$\begin{aligned} f_0(x^\star) &= g(\lambda^\star, \nu^\star) \\ &\leq f_0(x^\star) + \sum_{i=1}^m {\lambda_i^\star}^T f_i(x^\star) + \sum_{i=1}^p \nu_i^\star h_i(x^\star) \\ &\leq f_0(x^\star), \end{aligned}$$

and therefore we conclude that $x^\star$ minimizes $L(x, \lambda^\star, \nu^\star)$, and also that the two sums in the second line are zero. Since the second sum is zero (since $x^\star$ satisfies the equality constraints), we have $\sum_{i=1}^m {\lambda_i^\star}^T f_i(x^\star) = 0$. Since each term in this sum is nonpositive, we conclude that

$$ {\lambda_i^\star}^T f_i(x^\star) = 0, \quad i = 1, \ldots, m, \tag{5.94} $$

which generalizes the complementary slackness condition (5.48). From (5.94) we can conclude that

$$ \lambda_i^\star \succ_{K_i^*} 0 \implies f_i(x^\star) = 0, \qquad f_i(x^\star) \prec_{K_i} 0, \implies \lambda_i^\star = 0. $$

However, in contrast to problems with scalar inequalities, it is possible to satisfy (5.94) with $\lambda_i^\star \neq 0$ and $f_i(x^\star) \neq 0$.

KKT conditions

Now we add the assumption that the functions f_i, h_i are differentiable, and generalize the KKT conditions of §5.5.3 to problems with generalized inequalities. Since $x^\star$ minimizes $L(x, \lambda^\star, \nu^\star)$, its gradient with respect to x vanishes at $x^\star$:

$$ \nabla f_0(x^\star) + \sum_{i=1}^m Df_i(x^\star)^T \lambda_i^\star + \sum_{i=1}^p \nu_i^\star \nabla h_i(x^\star) = 0, $$

where $Df_i(x^\star) \in \mathbf{R}^{k_i \times n}$ is the derivative of f_i evaluated at $x^\star$ (see §A.4.1). Thus, if strong duality holds, any primal optimal $x^\star$ and any dual optimal $(\lambda^\star, \nu^\star)$ must satisfy the optimality conditions (or KKT conditions)

$$\begin{aligned} f_i(x^\star) &\preceq_{K_i} 0, \quad i = 1, \ldots, m \\ h_i(x^\star) &= 0, \quad i = 1, \ldots, p \\ \lambda_i^\star &\succeq_{K_i^*} 0, \quad i = 1, \ldots, m \\ {\lambda_i^\star}^T f_i(x^\star) &= 0, \quad i = 1, \ldots, m \\ \nabla f_0(x^\star) + \textstyle\sum_{i=1}^m Df_i(x^\star)^T \lambda_i^\star + \sum_{i=1}^p \nu_i^\star \nabla h_i(x^\star) &= 0. \end{aligned} \tag{5.95}$$

If the primal problem is convex, the converse also holds, *i.e.*, the conditions (5.95) are sufficient conditions for optimality of $x^\star$, $(\lambda^\star, \nu^\star)$.

5.9.3 Perturbation and sensitivity analysis

The results of §5.6 can be extended to problems involving generalized inequalities. We consider the associated perturbed version of the problem,

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \preceq_{K_i} u_i, \quad i = 1, \ldots, m \\ & h_i(x) = v_i, \quad i = 1, \ldots, p, \end{array}$$

where $u_i \in \mathbf{R}^{k_i}$, and $v \in \mathbf{R}^p$. We define $p^\star(u, v)$ as the optimal value of the perturbed problem. As in the case with scalar inequalities, $p^\star$ is a convex function when the original problem is convex.

Now let $(\lambda^\star, \nu^\star)$ be optimal for the dual of the original (unperturbed) problem, which we assume has zero duality gap. Then for all u and v we have

$$p^\star(u, v) \geq p^\star - \sum_{i=1}^{m} \lambda_i^{\star T} u_i - \nu^{\star T} v,$$

the analog of the global sensitivity inequality (5.57). The local sensitivity result holds as well: If $p^\star(u, v)$ is differentiable at $u = 0$, $v = 0$, then the optimal dual variables $\lambda_i^\star$ satisfies

$$\lambda_i^\star = -\nabla_{u_i} p^\star(0, 0),$$

the analog of (5.58).

Example 5.13 *Semidefinite program in inequality form.* We consider a semidefinite program in inequality form, as in example 5.11. The primal problem is

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & F(x) = x_1 F_1 + \cdots + x_n F_n + G \preceq 0, \end{array}$$

with variable $x \in \mathbf{R}^n$ (and $F_1, \ldots, F_n, G \in \mathbf{S}^k$), and the dual problem is

$$\begin{array}{ll} \text{maximize} & \mathbf{tr}(GZ) \\ \text{subject to} & \mathbf{tr}(F_i Z) + c_i = 0, \quad i = 1, \ldots, n \\ & Z \succeq 0, \end{array}$$

with variable $Z \in \mathbf{S}^k$.

Suppose that $x^\star$ and $Z^\star$ are primal and dual optimal, respectively, with zero duality gap. The complementary slackness condition is $\mathbf{tr}(F(x^\star)Z^\star) = 0$. Since $F(x^\star) \preceq 0$ and $Z^\star \succeq 0$, we can conclude that $F(x^\star)Z^\star = 0$. Thus, the complementary slackness condition can be expressed as

$$\mathcal{R}(F(x^\star)) \perp \mathcal{R}(Z^\star),$$

i.e., the ranges of the primal and dual matrices are orthogonal.

Let $p^\star(U)$ denote the optimal value of the perturbed SDP

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & F(x) = x_1 F_1 + \cdots + x_n F_n + G \preceq U. \end{array}$$

Then we have, for all U, $p^\star(U) \geq p^\star - \mathbf{tr}(Z^\star U)$. If $p^\star(U)$ is differentiable at $U = 0$, then we have

$$\nabla p^\star(0) = -Z^\star.$$

This means that for U small, the optimal value of the perturbed SDP is very close to (the lower bound) $p^\star - \mathbf{tr}(Z^\star U)$.

5.9.4 Theorems of alternatives

We can derive theorems of alternatives for systems of generalized inequalities and equalities

$$f_i(x) \preceq_{K_i} 0, \quad i = 1, \ldots, m, \qquad h_i(x) = 0, \quad i = 1, \ldots, p, \tag{5.96}$$

where $K_i \subseteq \mathbf{R}^{k_i}$ are proper cones. We will also consider systems with strict inequalities,

$$f_i(x) \prec_{K_i} 0, \quad i = 1, \ldots, m, \qquad h_i(x) = 0, \quad i = 1, \ldots, p. \tag{5.97}$$

We assume that $\mathcal{D} = \bigcap_{i=0}^m \mathbf{dom}\, f_i \cap \bigcap_{i=1}^p \mathbf{dom}\, h_i$ is nonempty.

Weak alternatives

We associate with the systems (5.96) and (5.97) the dual function

$$g(\lambda, \nu) = \inf_{x \in \mathcal{D}} \left(\sum_{i=1}^m \lambda_i^T f_i(x) + \sum_{i=1}^p \nu_i h_i(x) \right)$$

where $\lambda = (\lambda_1, \ldots, \lambda_m)$ with $\lambda_i \in \mathbf{R}^{k_i}$ and $\nu \in \mathbf{R}^p$. In analogy with (5.76), we claim that

$$\lambda_i \succeq_{K_i^*} 0, \quad i = 1, \ldots, m, \qquad g(\lambda, \nu) > 0 \tag{5.98}$$

is a weak alternative to the system (5.96). To verify this, suppose there exists an x satisfying (5.96) and (λ, ν) satisfying (5.98). Then we have a contradiction:

$$0 < g(\lambda, \nu) \leq \lambda_1^T f_1(x) + \cdots + \lambda_m^T f_m(x) + \nu_1 h_1(x) + \cdots + \nu_p h_p(x) \leq 0.$$

Therefore at least one of the two systems (5.96) and (5.98) must be infeasible, *i.e.*, the two systems are weak alternatives.

In a similar way, we can prove that (5.97) and the system

$$\lambda_i \succeq_{K_i^*} 0, \quad i = 1, \ldots, m, \qquad \lambda \neq 0, \qquad g(\lambda, \nu) \geq 0.$$

form a pair of weak alternatives.

Strong alternatives

We now assume that the functions f_i are K_i-convex, and the functions h_i are affine. We first consider a system with strict inequalities

$$f_i(x) \prec_{K_i} 0, \quad i = 1, \ldots, m, \qquad Ax = b, \tag{5.99}$$

and its alternative

$$\lambda_i \succeq_{K_i^*} 0, \quad i = 1, \ldots, m, \qquad \lambda \neq 0, \qquad g(\lambda, \nu) \geq 0. \tag{5.100}$$

We have already seen that (5.99) and (5.100) are weak alternatives. They are also strong alternatives provided the following constraint qualification holds: There exists an $\tilde{x} \in \mathbf{relint}\, \mathcal{D}$ with $A\tilde{x} = b$. To prove this, we select a set of vectors $e_i \succ_{K_i} 0$, and consider the problem

$$\begin{array}{ll} \text{minimize} & s \\ \text{subject to} & f_i(x) \preceq_{K_i} se_i, \quad i = 1, \ldots, m \\ & Ax = b \end{array} \tag{5.101}$$

with variables x and $s \in \mathbf{R}$. Slater's condition holds since $(\tilde{x}, \tilde{s})$ satisfies the strict inequalities $f_i(\tilde{x}) \prec_{K_i} \tilde{s}e_i$ provided $\tilde{s}$ is large enough.

The dual of (5.101) is

$$\begin{array}{ll} \text{maximize} & g(\lambda, \nu) \\ \text{subject to} & \lambda_i \succeq_{K_i^*} 0, \quad i = 1, \ldots, m \\ & \sum_{i=1}^m e_i^T \lambda_i = 1 \end{array} \tag{5.102}$$

with variables $\lambda = (\lambda_1, \ldots, \lambda_m)$ and ν.

Now suppose the system (5.99) is infeasible. Then the optimal value of (5.101) is nonnegative. Since Slater's condition is satisfied, we have strong duality and the dual optimum is attained. Therefore there exist $(\tilde{\lambda}, \tilde{\nu})$ that satisfy the constraints of (5.102) and $g(\tilde{\lambda}, \tilde{\nu}) \geq 0$, *i.e.*, the system (5.100) has a solution.

As we noted in the case of scalar inequalities, existence of an $x \in \mathbf{relint}\, \mathcal{D}$ with $Ax = b$ is not sufficient for the system of nonstrict inequalities

$$f_i(x) \preceq_{K_i} 0, \quad i = 1, \ldots, m, \qquad Ax = b$$

and its alternative

$$\lambda_i \succeq_{K_i^*} 0, \quad i = 1, \ldots, m, \qquad g(\lambda, \nu) > 0$$

to be strong alternatives. An additional condition is required, *e.g.*, that the optimal value of (5.101) is attained.

Example 5.14 *Feasibility of a linear matrix inequality.* The following systems are strong alternatives:

$$F(x) = x_1 F_1 + \cdots + x_n F_n + G \prec 0,$$

where $F_i, G \in \mathbf{S}^k$, and

$$Z \succeq 0, \qquad Z \neq 0, \qquad \mathbf{tr}(GZ) \geq 0, \qquad \mathbf{tr}(F_i Z) = 0, \quad i = 1, \ldots, n,$$

where $Z \in \mathbf{S}^k$. This follows from the general result, if we take for K the positive semidefinite cone $\mathbf{S}_+^k$, and

$$g(Z) = \inf_x \left(\mathbf{tr}(F(x)Z)\right) = \begin{cases} \mathbf{tr}(GZ) & \mathbf{tr}(F_i Z) = 0, \quad i = 1, \ldots, n \\ -\infty & \text{otherwise.} \end{cases}$$

The nonstrict inequality case is slightly more involved, and we need an extra assumption on the matrices F_i to have strong alternatives. One such condition is

$$\sum_{i=1}^{n} v_i F_i \succeq 0 \Longrightarrow \sum_{i=1}^{n} v_i F_i = 0.$$

If this condition holds, the following systems are strong alternatives:

$$F(x) = x_1 F_1 + \cdots + x_n F_n + G \preceq 0$$

and

$$Z \succeq 0, \qquad \mathbf{tr}(GZ) > 0, \qquad \mathbf{tr}(F_i Z) = 0, \quad i = 1, \ldots, n$$

(see exercise 5.44).

Bibliography

Lagrange duality is covered in detail by Luenberger [Lue69, chapter 8], Rockafellar [Roc70, part VI], Whittle [Whi71], Hiriart-Urruty and Lemaréchal [HUL93], and Bertsekas, Nedić, and Ozdaglar [Ber03]. The name is derived from Lagrange's method of multipliers for optimization problems with equality constraints; see Courant and Hilbert [CH53, chapter IV].

The max-min result for matrix games in §5.2.5 predates linear programming duality. It is proved via a theorem of alternatives by von Neuman and Morgenstern [vNM53, page 153]. The strong duality result for linear programming on page 227 is due to von Neumann [vN63] and Gale, Kuhn, and Tucker [GKT51]. Strong duality for the nonconvex quadratic problem (5.32) is a fundamental result in the literature on trust region methods for nonlinear optimization (Nocedal and Wright [NW99, page 78]). It is also related to the S-procedure in control theory, discussed in appendix §B.1. For an extension of the proof of strong duality of §5.3.2 to the refined Slater condition (5.27), see Rockafellar [Roc70, page 277].

Conditions that guarantee the saddle-point property (5.47) can be found in Rockafellar [Roc70, part VII] and Bertsekas, Nedić, and Ozdaglar [Ber03, chapter 2]; see also exercise 5.25.

The KKT conditions are named after Karush (whose unpublished 1939 Master's thesis is summarized in Kuhn [Kuh76]), Kuhn, and Tucker [KT51]. Related optimality conditions were also derived by John [Joh85]. The water-filling algorithm in example 5.2 has applications in information theory and communications (Cover and Thomas [CT91, page 252]).

Farkas' lemma was published by Farkas [Far02]. It is the best known theorem of alternatives for systems of linear inequalities and equalities, but many variants exist; see Mangasarian [Man94, §2.4]. The application of Farkas' lemma to asset pricing (example 5.10) is discussed by Bertsimas and Tsitsiklis [BT97, page 167] and Ross [Ros99].

The extension of Lagrange duality to problems with generalized inequalities appears in Isii [Isi64], Luenberger [Lue69, chapter 8], Berman [Ber73], and Rockafellar [Roc89, page 47]. It is discussed in the context of cone programming in Nesterov and Nemirovski [NN94, §4.2] and Ben-Tal and Nemirovski [BTN01, lecture 2]. Theorems of alternatives for generalized inequalities were studied by Ben-Israel [BI69], Berman and Ben-Israel [BBI71], and Craven and Kohila [CK77]. Bellman and Fan [BF63], Wolkowicz [Wol81], and Lasserre [Las95] give extensions of Farkas' lemma to linear matrix inequalities.